MODULE 4.2 · Audio and Speech Models

ASR: Automatic Speech Recognition Models

From HMM-GMM pipelines to Whisper and wav2vec — 50 years of speech recognition in one session

9 min	B2 – Understand	L2 – Intermediate	TH-000007
Duration	Bloom's	Level	Prerequisite

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Session Overview

What you will learn and why ASR is a foundational capability for voice AI

What You Will Learn

- What ASR is and how it works mathematically
- The evolution from HMM-GMM to deep learning ASR
- CTC loss: enabling alignment-free sequence training
- Attention-based encoder-decoder (Listen, Attend and Spell)
- Conformer: CNN + Transformer for state-of-the-art streaming ASR
- Whisper: large-scale multitask ASR
- wav2vec 2.0: self-supervised learning for low-resource ASR
- Word Error Rate (WER) and how to evaluate ASR systems

Concept at a Glance

- Duration: 9 minutes
- Bloom's Level: B2 – Understand
- Difficulty: L2 – Intermediate
- Module: 4.2 Audio and Speech Models
- Prerequisite: Spectrogram and Audio Features (TH-000007)
- Next: TTS Systems (TH-000009)

Learning Objectives

By the end of this session you will be able to:

9 min	B2 – Understand	L2 – Intermediate	TH-000007
Duration	Bloom's	Level	Prerequisite

1. Define Automatic Speech Recognition and state its formal mathematical objective
2. Explain the evolution from HMM-GMM pipelines to modern end-to-end deep learning ASR
3. Describe how CTC loss enables alignment-free sequence-to-sequence training
4. Identify the architectural components of Whisper, Conformer, and wav2vec 2.0
5. Calculate Word Error Rate (WER) and interpret its significance and limitations
6. Select the appropriate ASR model for a given production use case balancing accuracy, latency, and cost

SECTION 1 — THE ASR PROBLEM

What Is Automatic Speech Recognition?

Turning spoken words into text — formally and practically

The ASR Problem: Formal Definition

ASR is a sequence-to-sequence problem: find the most likely word sequence given audio observations

$$W^* = \arg\max_W P(W \mid X) = \arg\max_W P(X \mid W) \cdot P(W)$$

W = optimal word sequence | X = audio features | P(X|W) = acoustic model | P(W) = language model*

Variability

Same word sounds different across speakers, accents, speaking rates, recording conditions. A model must generalise across millions of voice patterns.

Coarticulation

Sounds blend into each other at natural speaking speed. No clear acoustic boundaries between phonemes — /b/ in 'bad' ≠ /b/ in 'lab'.

Noise

Background music, other speakers, room reverberation overlap with target speech. Real-world WER is 2–5× higher than clean-audio benchmarks.

Long-Range Context

Disambiguating homophones ('bank' financial vs river) requires understanding the full sentence. Local models systematically fail on pragmatic language.

Output options: Plain text, time-aligned words (with timestamps per word), or speaker-diarised segments (who said what when). Modern systems like Whisper produce all three simultaneously from a single forward pass.

SECTION 2 — EVOLUTION OF ASR

50 Years of ASR: A Visual Timeline

From hand-engineered phoneme models to large-scale neural sequence modelling

The ASR Timeline: Five Eras

Each era brought a fundamental shift in how the problem was approached

1970–2010

HMM-GMM Pipelines

Hand-crafted phoneme HMMs
Gaussian Mixture acoustic models
Pronunciation lexicon
Viterbi decoder

Kaldi, CMU Sphinx, HTK

2012–2016

HMM-DNN Hybrid

DNN replaces GMM as acoustic emission model
HMM decoding retained
Deep features from spectrograms

Deep Speech (Baidu)
Microsoft DNN-HMM

2016–2019

CTC / Attention End-to-End

No phoneme labels required
CTC loss for alignment-free training
Attention-based LAS decoder
Joint acoustic+language model

Listen, Attend and Spell
Deep Speech 2

2020–2022

Transformer Conformer

Self-attention for global context
CNN module for local patterns
Conformer = CNN + Transformer
wav2vec 2.0 self-supervised SSL

Conformer, wav2vec 2.0
HuBERT, WavLM

2022–now

Whisper / Native Audio LLMs

680K hrs weak supervision
Multilingual + multitask
GPT-4o native audio tokens
ASR as language modelling

Whisper, Gemini Audio
GPT-4o, SeamlessM4T

Key trend: Each era: more end-to-end, less hand-engineering, more data. Today ASR is a special case of large-scale sequence modelling — the same Transformer architecture powers GPT, Whisper, and Gemini Audio.

SECTION 3 — CTC LOSS

Connectionist Temporal Classification (CTC)

The algorithm that enabled end-to-end ASR without frame-level alignment labels

CTC: The Alignment Problem and Its Solution

Audio has 500 frames for 5 seconds; the transcript has 20 characters — how do you train without knowing which frames map to which characters?

The Alignment Problem

Audio: 500 frames | Transcript: 20 characters | No frame-level labels | **Which frames correspond to which characters?**

CTC Solution

1. Introduce a special <blank> token Ø
2. Output one token per frame (including blanks)
3. Sum over ALL valid alignments that collapse to the target
4. Collapse rule: merge repeated tokens, remove blanks

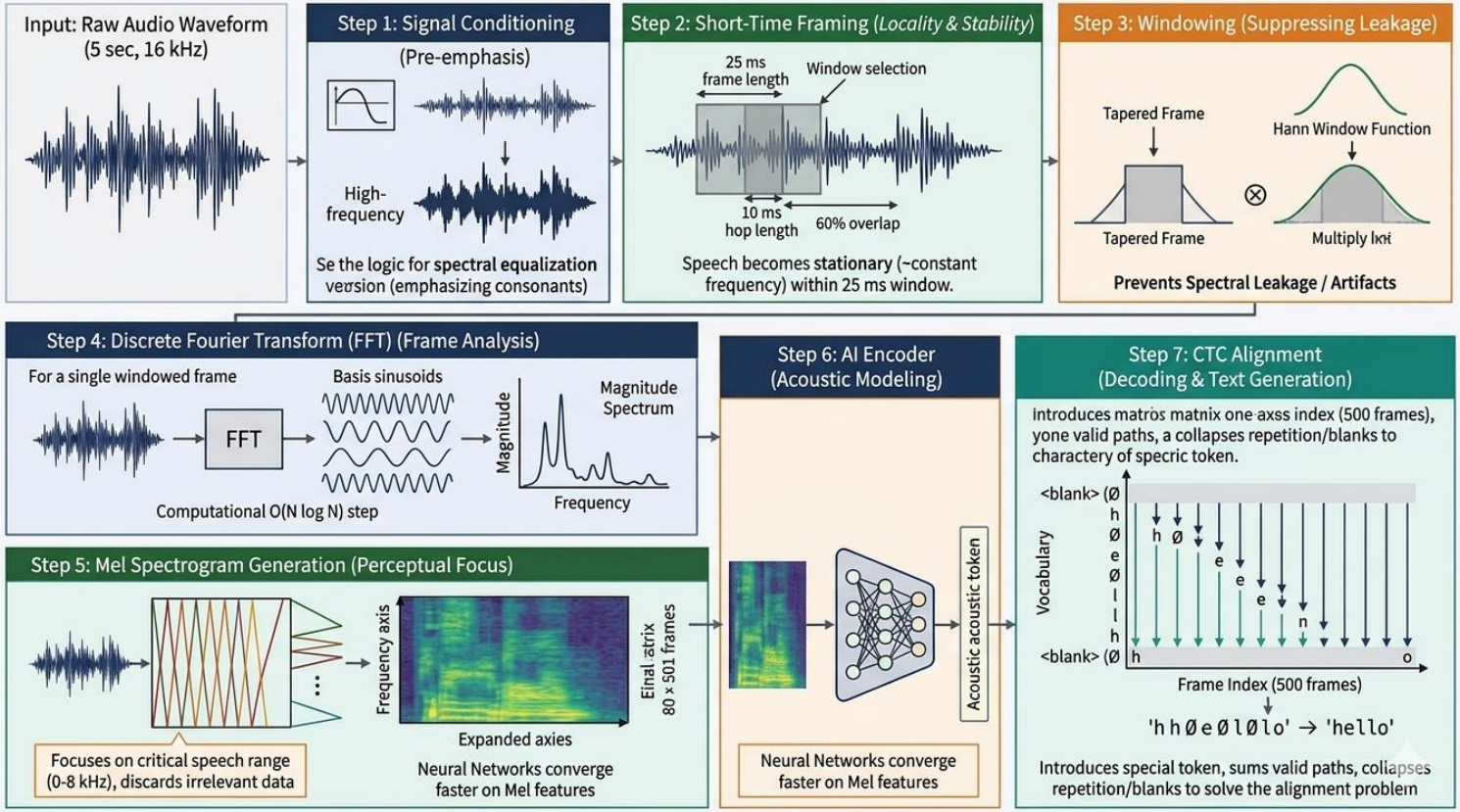
'h h Ø e Ø l Ø l o' → 'h e l l o'

Key Properties

- Conditional independence: output at each frame independent given encoder → efficient Viterbi decoding
- No need for frame-level phoneme annotations — only text transcripts required
- Streaming-compatible: output tokens incrementally as audio arrives
- Limitation: cannot model output-side language context without a separate LM

Modern use: CTC remains the backbone of streaming ASR (Conformer-Transducer, RNN-T). Google Assistant on Android, YouTube live captions, and Zoom use CTC-based streaming models. Non-streaming (batch) systems like Whisper prefer attention-based decoding.

CTC: The Alignment Problem and Its Solution



SECTION 4 — CONFORMER AND WHISPER

Modern ASR Architectures

The two dominant architectures: Conformer for streaming, Whisper for offline multitask

Conformer: CNN + Transformer for the Best of Both Worlds

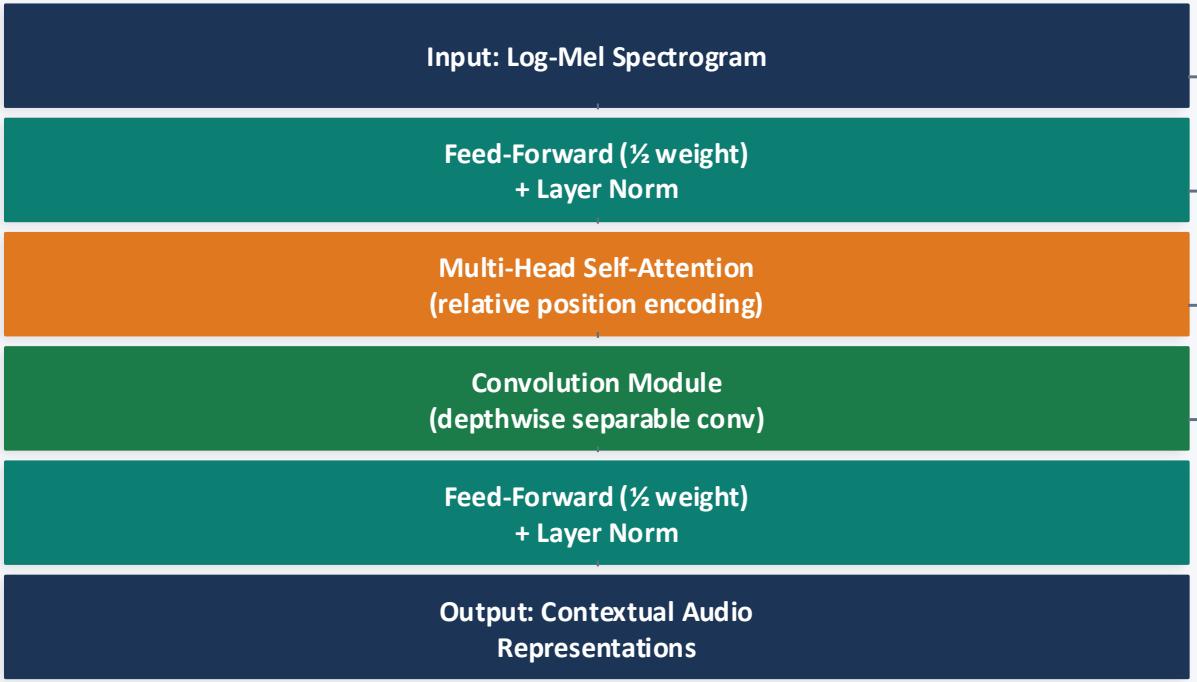
Pure Transformers excel at long-range dependencies; CNNs excel at local acoustic patterns. Conformer combines both.

Conformer Block (stack 12–18×)

Google, 2020

15.9% relative WER improvement vs pure Transformer on LibriSpeech

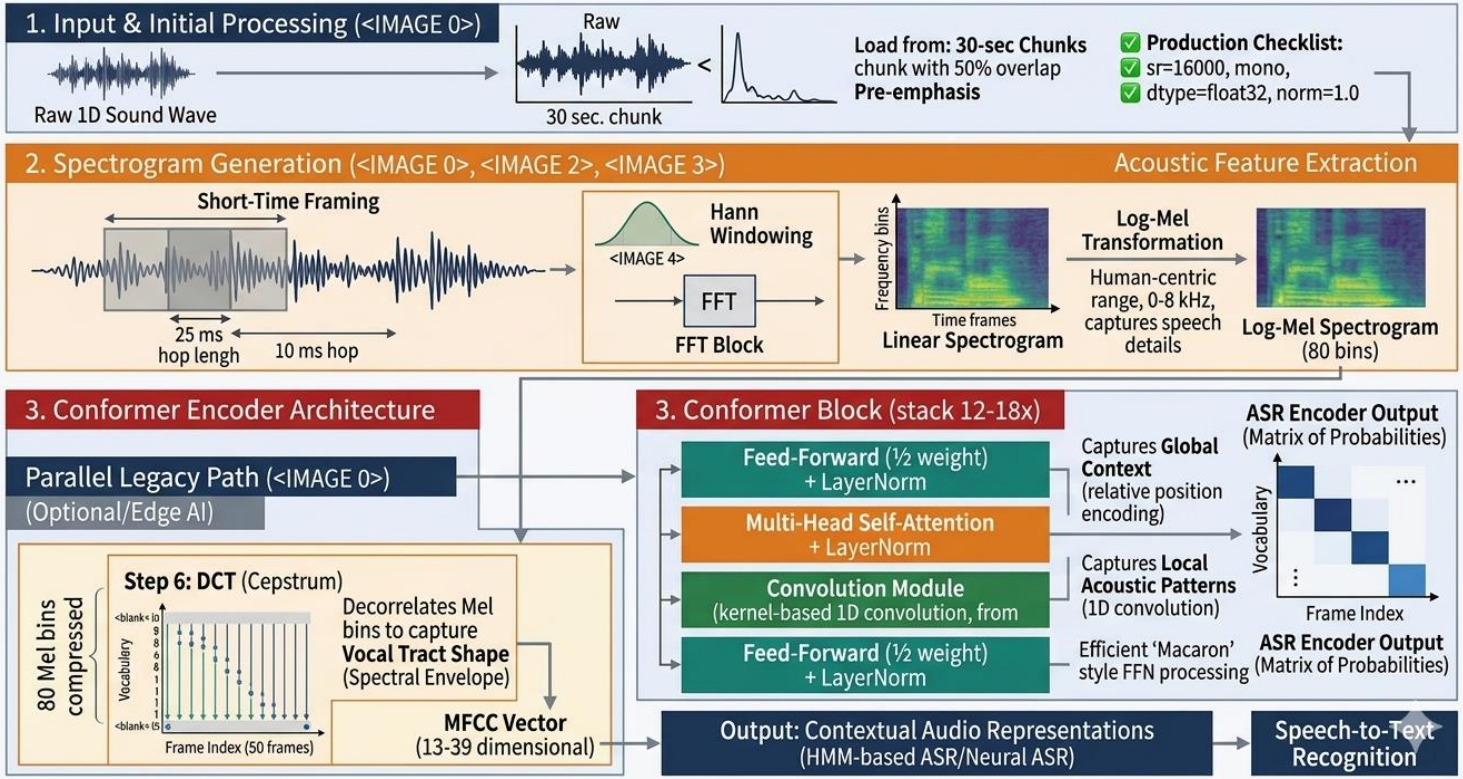
Used by:
- Google Assistant
- Android on-device ASR
- YouTube captions



Streaming variant: Conformer-Transducer (Conformer encoder + RNN-T prediction + joiner). Outputs tokens incrementally as each audio frame arrives. Latency: 200–300 ms. Google, YouTube, Zoom all use this in production.

Conformer: CNN + Transformer for the Best of Both Worlds

Pure Transformers excel at long-range dependencies; CNNs excel at local acoustic patterns. Conformer combines both.



Whisper: Large-Scale Multitask ASR

Radford et al. (OpenAI, 2022) — 680,000 hours of weakly-supervised web audio changed everything

Training Data

680,000 hours of web audio in 99 languages — 100× more than any prior single-dataset approach. Labels are weakly supervised (scraped transcripts, not human-verified).

Architecture

Standard encoder-decoder Transformer. Encoder: 2×Conv1D + Transformer blocks processes 80-bin log-Mel. Decoder: autoregressive Transformer with cross-attention to encoder output.

Multitask Capability

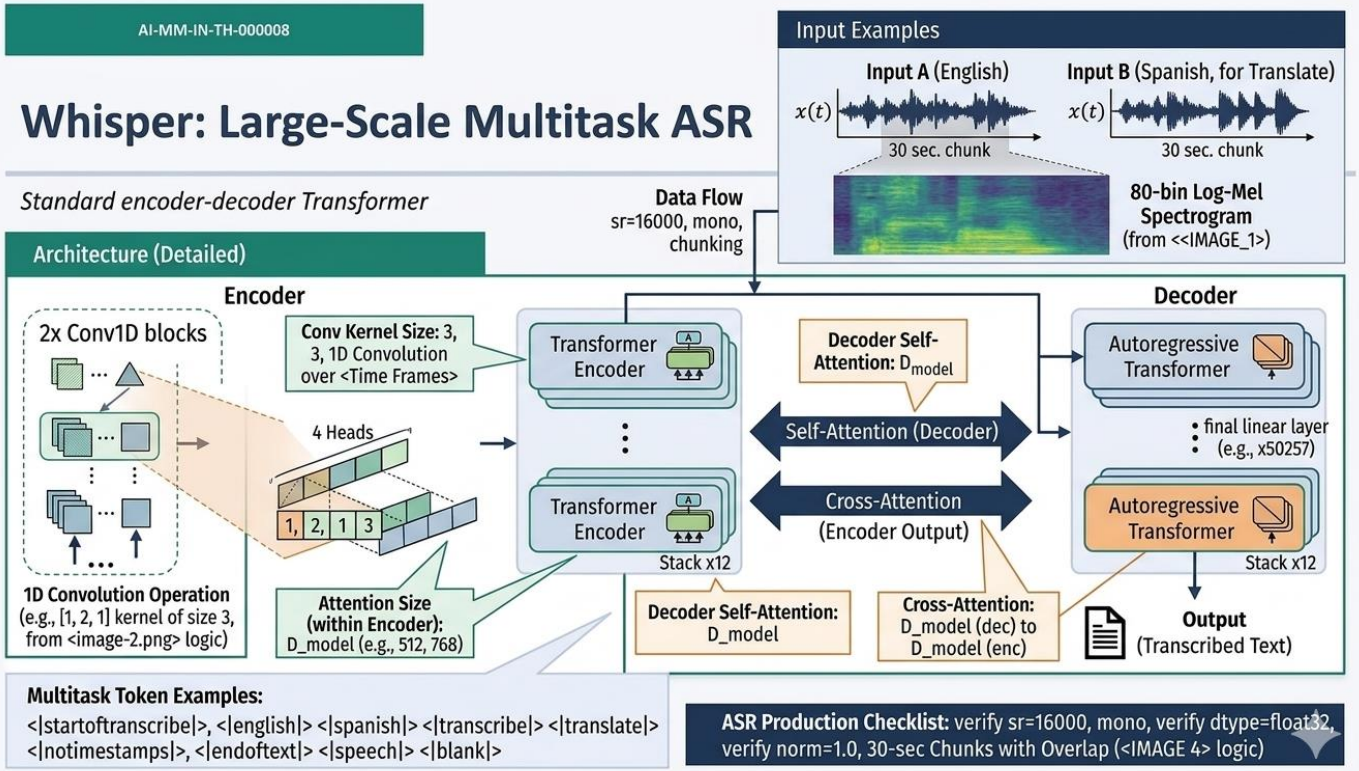
Single model handles: transcription, translation to English, language identification, voice activity detection, timestamp prediction. All controlled by special task tokens prepended to the decoder.

Performance

Whisper Large-v3: 2.7% WER on LibriSpeech test-clean. Matches specialist fine-tuned models without any fine-tuning. Available as API (OpenAI) or self-hosted (faster-whisper, 4–8× faster).

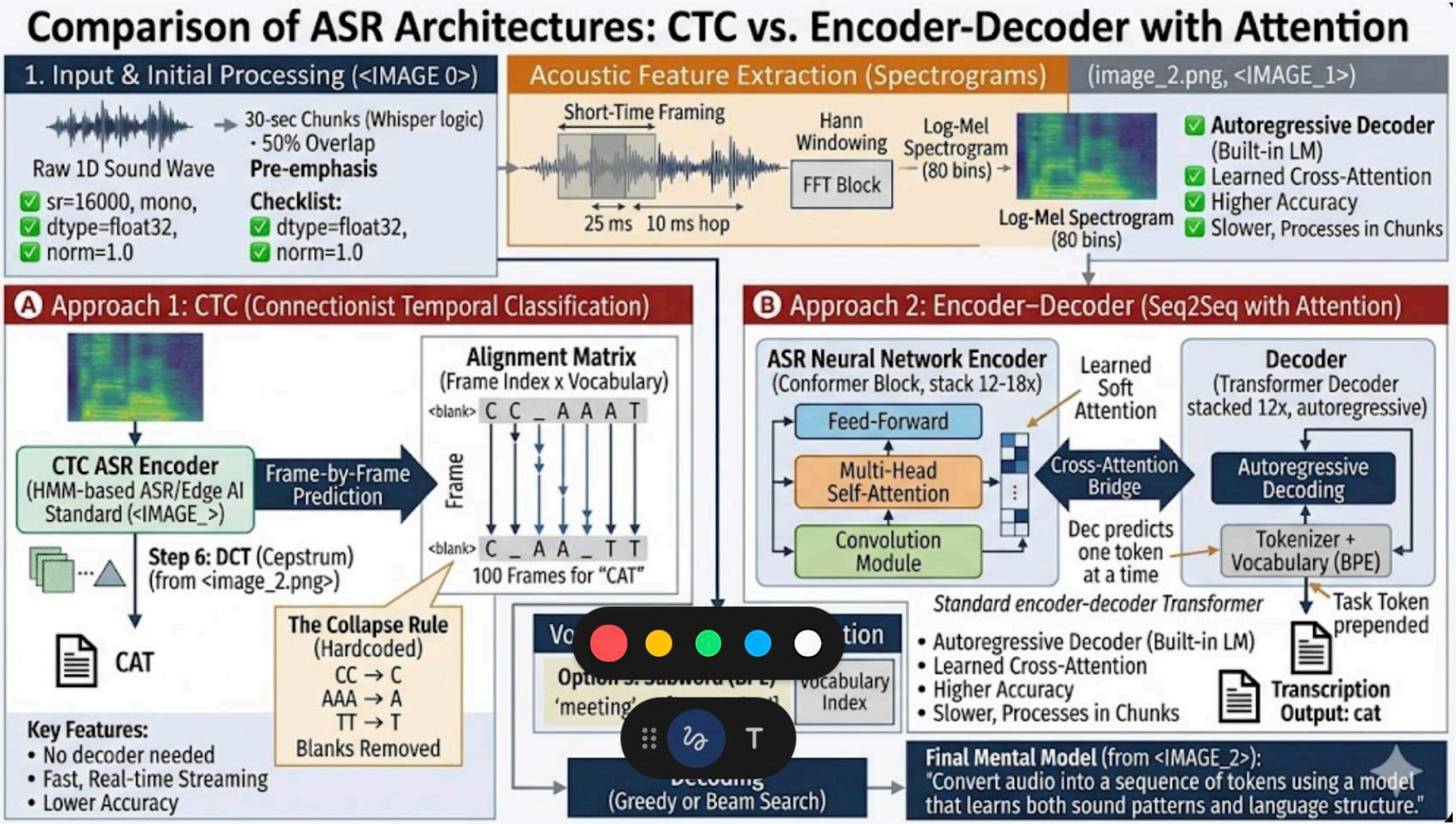
Key insight: Whisper demonstrated that data scale with weak supervision can match carefully curated supervised training. This fundamentally changed how ASR models are trained and deployed — you can now get near-human performance without a single human-verified label.

Audio → Spectrogram → Conv Layers → Transformer Encoder → Transformer Decoder → Text



Each spectrogram frame acts like a token, but the actual token embeddings are learned by the Conv layers (not a lookup table), and positional encoding is still used — just added after the Conv stage instead of at the raw input.

Audio → Spectrogram → Conv Layers → Transformer Encoder → Transformer Decoder → Text



SECTION 5 — SELF-SUPERVISED ASR

wav2vec 2.0 and Self-Supervised Learning

Achieving near-human WER with 10 minutes of labeled data — the low-resource revolution

wav2vec 2.0: Learning from Unlabeled Audio

Baevski et al. (Meta AI, 2020) — the key idea: learn from 960 hours unlabeled, fine-tune with 10 minutes labeled

Architecture	Training	Result
CNN feature extractor	Raw waveform → latent representations (local, ~20 ms window)	Strong local acoustic features no Mel preprocessing needed
Quantisation module	Latent reps → discrete codes via product quantisation	Converts continuous to discrete (like tokenisation for audio)
Transformer context network	Processes full context bidirectionally via self-attention	Global context modelling over all audio positions
Contrastive loss	Predict correct quantised code for masked frames vs distractors	Self-supervised — no labels needed during 960h pretraining

Results: Fine-tuned with 10 min labeled speech → <5% WER on LibriSpeech. 1 hour → 2.9% WER. Full 960 hrs → 1.8% WER. Why this matters: 99% of world languages have no large labeled speech dataset. wav2vec enables ASR for low-resource languages.

Evaluating ASR: Word Error Rate (WER)

The standard metric for ASR quality — computed as the edit distance between reference and hypothesis at the word level

$$\text{WER} = (S + D + I) / N \times 100\%$$

S = substitutions → wrong word | D = deletions→ missing word | I = insertions→ extra word | N = total words in reference

Worked Example

Reference: "The quick brown fox"
Hypothesis: "The thick brown fix the"

S=2 (quick→thick, fox→fix), D=0, I=1 (extra 'the')
WER = (2+0+1) / 4 × 100% = 75%

Benchmark WER Values

Human parity (LibriSpeech clean): ~2.5–5.5%

Whisper Large-v3 (test-clean): 2.7% WER

Conformer-CTC L (test-clean): 1.7% WER

Real-world (noisy, accented): 2–5× higher than benchmark

WER limitations: WER does not distinguish semantic errors ('one' vs 'won'). CER (Character Error Rate) is used for Chinese/Japanese. BLEU is used for speech translation tasks. Also measure: RTF (Real-Time Factor), latency, memory footprint.

ASR Model Comparison: Choosing for Production

The right model depends on your latency, language, and data requirements

Model	Params	WER (LibriSpeech clean)	Languages	Best Use Case
Whisper Large-v3	1.55B	2.7%	99	Offline, multilingual, zero-shot
Conformer-CTC L	610M	1.7%	English	Enterprise streaming ASR
wav2vec 2.0 XL	600M	1.8%	English	Low-resource fine-tuning
WavLM Large	316M	1.8%	English	SUPERB benchmark tasks
MMS (Meta)	1B	~10%	1,100 langs	Low-resource, multilingual
Parakeet-RNNT (NVIDIA)	1.1B	1.7%	English	NVIDIA GPU streaming

Decision rule: Use Whisper for out-of-box multilingual transcription (no fine-tuning). Use wav2vec/WavLM + fine-tuning for your specific domain with <10 hrs labeled data. Use Conformer-Transducer for real-time streaming (<300 ms latency).

Key Takeaways

The six essential points from this concept

ASR solves W* = arg max P(W|X) Find the word sequence that maximises the posterior probability given audio, using Bayes theorem to decompose into acoustic model × language model.

50 years: HMM-GMM → end-to-end DL Each era eliminated more hand-engineering. Today's SOTA treats ASR as sequence modelling — same architecture as GPT for text.

CTC enables alignment-free training Blank token + marginalize over all valid alignments. No frame-level labels needed. Essential for streaming ASR (Conformer-Transducer).

Conformer = CNN + Transformer Local acoustic patterns (CNN) + global context (Transformer). Industry standard for production streaming ASR. 15.9% relative WER improvement vs pure Transformer.

Whisper: weak supervision at scale 680K hours + weak labels = near-human WER on 99 languages with zero fine-tuning. Data scale trumps label quality.

wav2vec 2.0: self-supervised for low-resource 10 minutes of labeled data → <5% WER. Self-supervised pre-training enables ASR for the 90% of

Note: Next: AI-MM-IN-TH-000009 — TTS Systems (Tacotron 2, FastSpeech 2, VALL-E, voice cloning)

AI-MM-IN-TH-000008 Complete

ASR: Automatic Speech Recognition Models

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